

Notes: Vestman (RFS, 2019)

Limited Stock Market Participation Among Renters and Homeowners

Contents

1 Big picture

2 Data and measurement (key objects)

2.1	Swedish panel data	
2.2	Household head and sample restrictions	
2.3	Key balance-sheet variables	
2.4	Descriptive facts	
2.5	Life-cycle profiles	

3 Model and empirical specifications (all equations + interpretation)

3.1	Demographics and household types	
3.2	Consumption basket and preferences	
3.3	Income process	
3.4	One-time stock market entry cost	
3.5	Asset returns and housing prices	
3.6	Borrowing, mortgages, and tenure choice	
3.7	Law of motion for cash-in-hand	
3.8	Decision rules and mechanism	
3.9	Calibration	
3.10	Difference-in-difference event-study specification	

4 Identification and credibility

4.1	What identifies the empirical estimates	
4.2	Why household fixed effects matter	
4.3	Pre-trends and event-study logic	
4.4	Household composition as the main confound	
4.5	Why the structural model is useful	
4.6	Limits and tensions	
4.7	Robustness	

5 Tables and figures

5.1	Table 2: Swedish descriptive statistics	
5.2	Table 3: Calibration targets and parameter values	
5.3	Figure 2: Life-cycle profiles for renters and homeowners	
5.4	Figure 5: Policy rules for type 1 versus type 2	

5.5	Table 4: DID estimates in the data	
5.6	Figure 7: Model fit for homeownership and conditional wealth profiles	
5.7	Figure 8: Stock market participation in model and data	
5.8	Table 5: DID estimates in the model	
5.9	Table 6: Other asset classes in the data	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

Question. Why is stock market participation much lower among renters than among homeowners, even though homeowners already have substantial wealth tied up in housing? Does homeownership *crowd out* stock market participation, or are renters and homeowners different types of households to begin with?

Core empirical fact. In the Swedish data, the stock market participation rate is roughly twice as high among homeowners as among renters. In the calibration sample with household heads who have high school education, participation is about 79% for homeowners and 43.5% for renters.

The puzzle. For two otherwise identical households, one renter and one homeowner, the renter should have *more* incentive to participate in the stock market because the renter has less wealth tied up in housing. So a simple crowding-out logic predicts the wrong cross-sectional ranking.

Main idea. The paper combines a life-cycle portfolio choice model with endogenous housing tenure choice and a stylized form of preference heterogeneity. The high-savings type tends to accumulate more financial wealth, buy homes, and pay the entry cost to participate in the stock market. The low-savings type tends to remain poorer, rent more often, and skip participation.

Main theoretical contribution. The paper shows that once tenure choice is endogenous, cross-sectional differences in stock market participation can reflect *sorting* rather than a pure treatment effect of homeownership. Homeownership can crowd out participation for otherwise identical households, yet homeowners can still participate more than renters in equilibrium because the kinds of households who become homeowners are different.

Main quantitative result. A model with only modest heterogeneity and relatively small one-time participation costs — SEK 5,000 for the high-savings type and SEK 15,000 for the low-savings type — generates a large homeowner-renter participation gap. During working life, the model produces a gap of about 40 percentage points, close to the data.

Main empirical result. Difference-in-difference event-study regressions on Swedish panel data do *not* find evidence that becoming a homeowner reduces stock market participation. Home purchases are associated with weakly positive participation responses; home sales are associated with negative participation responses.

Main mechanism. Preference heterogeneity creates dispersion in financial wealth. That dispersion drives both tenure choice and stock market entry. Homeowners hold more equity risk not because housing mechanically pushes them into the stock market, but because the households who choose homeownership also tend to have stronger savings motives.

Main limitation. The model matches the participation gap well, but it overpredicts young homeownership and late-life stock market participation. On the intensive margin, the model predicts stronger equity-share responses to housing changes than the data support.

Economic takeaway. The paper weakens the standard “participation puzzle” by showing that housing tenure choice and moderate heterogeneity in preferences explain a large part of why so many renters stay out of the stock market.

2 Data and measurement (key objects)

2.1 Swedish panel data

The empirical analysis uses a tailored Swedish registry-based panel from Statistics Sweden. The foundation is LINDA (Longitudinal INdividual DAta for Sweden), which covers about 320,000 households — roughly 7% of the Swedish population — with annual waves from 2000 to 2007.

The key value added is that LINDA is matched to:

- KURU, which provides detailed holdings of direct stocks, equity mutual funds, bond funds, bonds, money market funds, and bank accounts;
- the real estate registry, which makes it possible to identify homeowners and trace housing upgrades and downgrades.

Interpretation. The paper has unusually strong micro data for this question because it observes both sides of household balance sheets: financial portfolios and housing positions.

2.2 Household head and sample restrictions

A household head is defined as the oldest male aged at least 21 if such a male exists; otherwise the oldest female aged at least 21; otherwise the oldest person in the household.

Two important sample restrictions are imposed:

- the top 1% of households by net worth are excluded, corresponding to roughly SEK 9 million in 2005 prices;
- for calibration and direct model-data comparisons, the paper focuses on households whose head has *high school* education but no college degree.

Interpretation. The second restriction is not the main result; it is a way to limit a major observable source of heterogeneity and keep the income process manageable.

2.3 Key balance-sheet variables

The main asset objects are:

- **Financial wealth:** direct stocks, equity mutual funds, bonds, money market funds, bank accounts, capital insurance, and private pension accounts.
- **Housing wealth:** the market value of houses, cabins, land, and co-op apartments.
- **Net worth:** financial wealth plus housing wealth minus debt.
- **Home equity:** net worth minus financial wealth.

The main portfolio outcome is **stock market participation**, defined as holding either direct stocks or equity mutual funds in liquid savings accounts.

The paper also studies:

- the **unconditional equity share**, coded as zero for nonparticipants;
- the **conditional equity share**, among participants only.

Interpretation. The extensive margin — whether a household enters the stock market at all — is the centerpiece. The intensive margin matters, but it plays a secondary role in the paper.

2.4 Descriptive facts

In the full Swedish sample:

- overall stock market participation is 63.3%;
- homeowners have participation 77.7%;
- renters have participation 37.7%.

In the high-school calibration sample:

- renters' participation is 43.5%;
- homeowners' participation is 79.1%;
- the gap is 35.6 percentage points.

Homeowners are also much richer. In the calibration sample, average real estate wealth is about SEK 1.35 million for homeowners and zero for renters, while average net worth is about SEK 1.07 million for homeowners and slightly negative for renters.

Interpretation. The empirical puzzle is not just that homeowners hold homes. It is that homeowners also hold *more* risky financial wealth, despite having more wealth tied up in housing.

2.5 Life-cycle profiles

The paper documents four basic life-cycle facts:

- disposable income is hump-shaped;
- homeownership rises over the life cycle and declines only mildly late in life;
- financial wealth peaks relatively late;
- stock market participation rises until around age 40 and is then fairly flat.

Conditional on tenure, homeowners have:

- higher disposable income;
- much higher financial wealth;
- a participation rate that is close to 80% from around age 30 onward.

Interpretation. These profiles suggest that the renter-homeowner participation gap is deeply intertwined with life-cycle wealth accumulation.

3 Model and empirical specifications (all equations + interpretation)

3.1 Demographics and household types

A household lives from age 25 to at most 95. One model period equals two years. Retirement occurs at age 65. There are two ex ante household types:

- **Type 1:** high-savings households;
- **Type 2:** low-savings households.

Their population weights are θ and $1 - \theta$, with the baseline setting $\theta = 0.75$.

Interpretation. The model does not assume observable heterogeneity like education or marital status drives everything. Instead, it uses a simple latent two-type structure to generate realistic wealth dispersion.

3.2 Consumption basket and preferences

The consumption basket is Cobb–Douglas:

$$C_t \equiv c_t^{1-\omega} h_t^\omega,$$

where c_t is non-housing consumption, h_t is housing services, and ω is the housing expenditure share.

Households have Epstein–Zin preferences:

$$U_t = \left[C_t^{1-\rho_j} + \beta_j(1 - \pi_t)R_t(U_{t+1})^{1-\rho_j} + \beta_j\pi_t b_j R_t(a_{f,j}X_{t+1})^{1-\rho_j} \right]^{\frac{1}{1-\rho_j}},$$

with certainty equivalent

$$R_t(U_{t+1}) = \mathbb{E}_t \left[U_{t+1}^{1-\gamma_j} \right]^{\frac{1}{1-\gamma_j}}.$$

Here:

- ρ_j governs the elasticity of intertemporal substitution;
- γ_j is relative risk aversion;
- β_j is the discount factor;
- π_t is the mortality probability;
- b_j , $a_{f,j}$, and $T_{b,j}$ govern accidental bequests.

Interpretation. This preference structure is what lets the model separate savings motives from risk attitudes. That flexibility is essential for matching both participation and the cross-sectional wealth distribution.

3.3 Income process

Labor income before retirement follows:

$$y_{it} \equiv \log Y_{it} = g_t + z_{it}, \quad t \leq 65,$$

$$z_{it} = z_{it-1} + \nu_{it}^o + \varepsilon_t^o + \eta_t^o, \quad t \leq 65,$$

and retirement income is

$$Y_{it} = \lambda \cdot \exp(g_{65} + z_{i65}), \quad t > 65.$$

The process contains:

- an age profile g_t ;
- idiosyncratic permanent shocks ν_{it}^o ;
- shocks correlated with the stock market ε_t^o ;
- shocks correlated with the housing market η_t^o .

The Swedish welfare system is modeled with an income floor $\underline{Y}$ and a cap $\bar{Y}$ on income counted toward pension replacement.

Interpretation. Income risk matters both directly, through precautionary saving, and indirectly, because its correlation with stock and housing returns affects optimal portfolio choice.

3.4 One-time stock market entry cost

To participate in the stock market, a household must pay a one-time cost κ_j . The participation state evolves as

$$I_t = \begin{cases} 1 & \text{if } I_{t-1} = 1 \text{ or } \alpha_t > 0, \\ 0 & \text{otherwise,} \end{cases} \quad I_{25} = 0,$$

where α_t is the fraction of financial wealth invested in equity.

Interpretation. The entry cost is what produces nonparticipation. The housing side of the model matters because tenure choice changes how much financial wealth households accumulate and therefore whether paying κ_j is worthwhile.

3.5 Asset returns and housing prices

The stock return and housing return follow:

$$R_{t+1} = \exp(\log(R_f) + \mu + \varepsilon_{t+1}),$$

$$R_{t+1}^h = \frac{P_{t+1}^h}{P_t^h} = \exp(\mu^h + \eta_{t+1} + \varepsilon_{t+1}^h).$$

The paper assumes:

- a constant risk-free return R_f ;
- an equity premium μ ;
- house-price growth μ^h ;
- correlation between labor income, stock returns, and housing returns.

Interpretation. Housing is both a consumption good and a risky asset. That dual role is central: housing affects the household's static budget and its future risk exposure.

3.6 Borrowing, mortgages, and tenure choice

The model imposes:

$$A_t \geq 0, \quad \alpha_t \in [0, 1],$$

and mortgage borrowing up to

$$0 \leq M_t \leq (1 - \delta)P_t^h h_t.$$

For tractability, the household is assumed to rebalance each period so that

$$M_t = (1 - \delta)P_t^h h_t.$$

If the household rents:

$$A_t + c_t + \tau P_t^h h_t \leq X_t.$$

If it buys a home:

$$A_t + c_t + (\phi_b + \chi + \delta)P_t^h h_t \leq X_t.$$

If it already owns and stays:

$$A_t + c_t + (\chi + \delta - \phi)P_t^h h_t \leq X_t.$$

Here:

- δ is the down payment;
- ϕ_b is the purchase cost;
- ϕ is the selling cost;
- χ is maintenance;
- τ is the rent-to-price ratio.

Interpretation. Renting and owning differ not just in current expenditure but in how wealth is stored and how risk is carried into the future.

3.7 Law of motion for cash-in-hand

If the household rents between t and $t + 1$:

$$X_{t+1} = A_t R_f + \alpha_t A_t (R_{t+1} - R_f) + Y_{t+1} - \kappa(I_{t+1} - I_t).$$

If it owns a home h_t :

$$X_{t+1} = A_t R_f + \alpha_t A_t (R_{t+1} - R_f) + Y_{t+1} - \kappa(I_{t+1} - I_t) + P_t^h h_t [R_{t+1}^h (1 - \phi) - (1 - \delta)R_f].$$

Interpretation. Homeownership directly changes next period cash-in-hand through capital gains or losses in housing. So tenure choice changes both wealth accumulation and risk exposure.

3.8 Decision rules and mechanism

The household solves dynamic programming problems corresponding to renting, buying, or staying in the current home.

The qualitative policy functions are central:

- for a *given* preference type, renters save more in financial assets than homeowners;
- therefore renters enter the stock market at *lower* cash-on-hand thresholds than homeowners;
- homeowners choose higher conditional equity shares because more wealth is tied up in housing.

Interpretation. This is the model’s “crowding-out” force. Holding the household fixed, homeownership makes stock participation less attractive. This force alone predicts the wrong cross-sectional ranking.

Once type heterogeneity is added:

- Type 1 households save more and are more likely to buy homes and enter the stock market;
- Type 2 households save less, consume more, and often remain renters and nonparticipants.

Interpretation. The homeowner-renter gap in participation is driven by *who becomes a homeowner*, not by a direct positive effect of housing on participation.

3.9 Calibration

The paper groups parameters into returns and risk, welfare and moving shocks, housing-market parameters, and preference/type-specific parameters.

Key calibrated values are:

- $R_f = 1\%$, equity premium = 4%, expected house-price growth = 1%, stock return volatility = 0.17;
- down payment $\delta = 0.30$, purchase cost $\phi_b = 0.10$, sale cost $\phi = 0.02$, rent-to-price ratio $\tau = 0.05$, minimum house value = 500 kSEK, maintenance $\chi = 0.043$;
- type weights: $\theta = 0.75$ for type 1;
- participation costs: $\kappa_1 = 5$ kSEK and $\kappa_2 = 15$ kSEK;
- type 1 preferences: $\gamma_1 = 5$, $1/\rho_1 = 1/3$;
- type 2 preferences: $\gamma_2 = \rho_2 = 1.2$;
- common discount factor: $\beta_1 = \beta_2 = 0.91$;
- housing share: $\omega = 0.25$;
- replacement rate in retirement: $\lambda = 0.65$.

Simulation uses 1,500 type-1 households and 500 type-2 households. Aggregate shocks are held at expected values so the simulated cross-section matches the empirical cross-section as closely as possible.

Interpretation. The model does not need implausibly large stock market entry costs once housing choice and preference heterogeneity are included.

3.10 Difference-in-difference event-study specification

For the panel-data part, the paper estimates an event-study DID regression around home purchases and home sales:

$$D_{it}(\alpha_{it} > 0) = \phi_0 + \text{Purchase}_i \times \sum_{k \neq -1} \delta_k RY_i(t = k) + \sum_{k \neq -1} \gamma_k RY_i(t = k) + \psi_t + \omega_i + \varepsilon_{it}.$$

Here:

- $D_{it}(\alpha_{it} > 0)$ is a participation dummy;
- $RY_i(t = k)$ are relative-year indicators;
- ψ_t are year fixed effects;
- ω_i are household fixed effects;
- δ_k are the event-time treatment effects.

The omitted relative year is $k = -1$, so testing $\delta_{-2} = 0$ is the pre-trend test.

The same framework is applied to:

- home purchases and home sales;
- unconditional equity shares;
- conditional equity shares;
- broader asset classes.

Interpretation. The empirical strategy is designed to separate permanent household heterogeneity from changes around housing transitions.

4 Identification and credibility

4.1 What identifies the empirical estimates

The DID regressions identify how portfolio outcomes change around home purchases or home sales *relative* to households that make no housing transaction. Identification comes from within-household changes over time, netting out household fixed effects and common year shocks.

Interpretation. This is much stronger than a pure cross-sectional comparison of renters and homeowners because renters may simply be different people.

4.2 Why household fixed effects matter

Household fixed effects absorb time-invariant characteristics such as:

- baseline patience or thriftiness;
- persistent risk tolerance;
- stable financial sophistication;
- persistent unobserved preferences for homeownership.

Interpretation. Without fixed effects, the participation gap could just reflect permanent types. The panel approach explicitly tries to remove that source of bias.

4.3 Pre-trends and event-study logic

The paper omits relative year -1 and checks whether $\delta_{-2} = 0$. If pre-trends are flat, then the post-event coefficients are more plausibly interpreted as responses to the housing transition.

Interpretation. The paper is careful not to overclaim. Even when pre-trends are flat, housing is not randomly assigned; the empirical design supports credibility, but not a literal randomized-treatment interpretation.

4.4 Household composition as the main confound

A major concern is that home transactions coincide with changes in household composition — especially marriage and divorce. Those events can affect risk tolerance, savings, and portfolio choice directly.

The paper addresses this by splitting the treatment and control groups into:

- stable households;
- single-to-married households;
- married-to-single households.

Interpretation. This is a key credibility step. It shows that the strongest negative responses around home sales are concentrated among households that simultaneously become single, which is exactly where a composition shock is likely to matter most.

4.5 Why the structural model is useful

The structural model is not identified off the DID regressions alone. Instead, it is disciplined by:

- life-cycle profiles of income, wealth, and participation;
- the cross-sectional distribution of financial wealth;
- homeownership profiles;
- the size of the homeowner-renter participation gap.

Interpretation. The model is used as a quantitative organizing device. It asks what kind of heterogeneity and what size of entry costs are needed to reproduce the joint housing-finance facts.

4.6 Limits and tensions

The paper is transparent about several shortcomings:

- the model overpredicts homeownership among the young;
- it overpredicts participation later in life;
- it predicts stronger equity-share responses to housing changes than the data reveal;
- it assumes frictionless mortgage rebalancing and a competitive rental market, both of which are stylized for Sweden.

Interpretation. The paper’s main message is strongest on the *extensive margin* of participation, weaker on the conditional equity-share margin.

4.7 Robustness

The paper studies several alternatives:

- **Bequest-motive heterogeneity:** does *not* generate enough wealth dispersion or a large participation gap;
- **Discount-factor heterogeneity:** works qualitatively, but the participation gap falls from about 40 to 20 percentage points during working life;
- **Cheaper rental market:** reduces the gap somewhat, but still leaves a substantial homeowner-renter participation difference;
- **No moving shocks (with higher down payment):** leaves the main patterns broadly intact.

Interpretation. The quantitative success does not rely on one single knife-edge assumption, but the *baseline* preference heterogeneity works best.

5 Tables and figures

5.1 Table 2: Swedish descriptive statistics

Read.

- Homeowners are much richer than renters in both housing wealth and net worth.
- Stock market participation is 77.7% for homeowners versus 37.7% for renters in the broad sample.
- In the high-school calibration sample, participation is 79.1% for homeowners and 43.5% for renters.
- The renter-homeowner gap is much larger than the gap across education groups.

Economic message. This table establishes the core fact the paper needs to explain: households with *more* wealth tied up in housing are also the ones who participate more in the stock market.

5.2 Table 3: Calibration targets and parameter values

Read.

- The model uses realistic financial and housing return moments.
- Housing is costly: a 30% down payment, 10% purchase cost, and 2% selling cost.
- The two types differ sharply in risk aversion and participation costs.
- Type 1 carries a strong precautionary-savings motive and low entry cost; type 2 has a weak savings motive and higher entry cost.

Economic message. The participation puzzle is reduced not by enormous stock-market frictions, but by combining housing choice with heterogeneity in savings motives.

Averages of variables: Swedish data

	All	Renters	Home-owners	Elem. school	High school	College	Renters ^{HS}	Home-owners ^{HS}
Equity mutual funds	79.1	27.4	108.4	64.9	66.3	112.0	20.6	88.1
Stocks	52.0	17.3	71.5	31.9	38.5	87.2	11.2	51.7
Bonds	23.5	10.0	31.2	23.9	18.0	25.1	6.4	23.6
Money market funds	12.9	7.3	16.0	12.3	9.9	13.6	4.7	12.3
Bank accounts	95.8	46.9	123.4	90.3	75.7	114.3	30.9	97.2
Participation	0.633	0.377	0.777	0.571	0.675	0.771	0.435	0.791
Equity share (uncond.)	0.370	0.242	0.442	0.318	0.410	0.458	0.297	0.464
Capital insurance	23.2	9.0	31.2	21.4	19.0	26.5	6.2	25.1
Pension accounts	46.6	10.9	66.7	42.0	44.1	74.5	12.6	59.2
Real estate	926.1	0	1,449	798.5	912.1	1,331	0	1,350
Net worth	820.0	27.2	1,268	761.7	714.6	1,086	-21.0	1,068
Loan-to-value	—	—	0.597	—	—	—	—	0.681
Disposable income	293.0	183.6	354.8	274.3	308.1	376.2	214.2	353.2
Age	47.7	43.5	50.1	51.6	43.9	43.9	39.7	45.8
Married (%)	47.7	26.6	59.6	53.5	46.2	53.2	25.4	56.2
Single (%)	52.3	73.4	40.4	46.5	53.8	46.8	74.6	43.8
Household size	2.58	2.02	2.90	2.58	2.84	2.84	2.29	3.10
Adults	1.53	1.15	1.75	1.64	1.59	1.61	1.21	1.76
Children	1.05	0.87	1.15	0.94	1.25	1.23	1.08	1.34
Move (%)	11.8	19.5	7.50	10.3	13.6	14.0	24.8	8.20
Observations	2,434,359	878,759	1,555,600	440,827	991,758	633,090	321,883	669,875

Asset values in terms of thousands of Swedish kronor (SEK) from 2000 to 2007, deflated to 2005 prices. Elementary school refers to 10 years of education or less of the household head. High school refers to 11 to 13 years of education, and college equals 14 or more years. Age refers to the age of the household head. The last two columns refer to renters and homeowners whose household heads have high school education. Stocks refers to direct holdings of stocks. Bonds refers to direct holdings and bond funds. The balance on private pension accounts was imputed using historical deposits since 1985. To obtain an estimate of the balance, the MSCI World Market return was applied to the deposits. Participation refers to the stock market participation rate in liquid savings accounts—that is, government-mandated premium pension accounts, capital insurances, and private pension accounts are excluded. Equity share is the value equity mutual funds and stock divided by liquid savings. It is coded to 0 if liquid savings is 0. Married is a dummy variable that equals 1 if the household head is married, otherwise 0. All non-married household heads are defined as singles. All household members older than 25 years are defined as adults, and all other members as children. Move is a dummy variable that equals 1 if the household head changed official address during the year. Columns 4–8 exclude in total 368,684 household-year observations for which there is no information about education (mainly older birth cohorts and immigrants).

participation rate among renters is 37.7% and among homeowners 77.7%, a gap of 40 percentage points. The fourth, fifth, and sixth columns split the sample by education. (If information about education is missing for the household head, the household is dropped. This mainly affects households with older heads and immigrant households.) There are big differences between college households and elementary school households in terms of disposable income

Table 3
Parameter values

	Notation	Value
Returns		
Risk-free rate	r_f	0.01
Equity premium	μ	0.04
Expected house price growth	μ_h	0.01
Standard deviation of stock market return	σ_e	0.17
Risks		
Standard dev. of idiosyncratic labor income shock	σ_v	0.1283
Standard dev. of stock market shock in labor income	σ_e^p	0.0130
Standard dev. of housing market shock in labor income	σ_h^p	0.0166
Standard dev. of housing market shock in stock market	σ_e^h	0.0150
Moving shocks and welfare system		
Probability of a move	ξ_t	—
Replacement rate in retirement	λ	0.65
Income floor	$\bar{y}$	60
Maximum income that counts toward pension	$\bar{y}$	506
Housing		
Down-payment	δ	0.30
Purchase cost	ϕ_b	0.10
Sale cost	ϕ	0.02
Rent to price ratio	τ	0.05
Minimum house value	$\bar{h}$	500
Maintenance	χ	0.043
Type 1: Preferences and stock market particip. cost		
Elasticity of intertemporal substitution	$1/\rho_1$	1/3
Relative risk aversion	γ_1	5
Stock market entry cost	κ_1	5
Type 2: Preferences and stock market particip. cost		
Elasticity of intertemporal substitution	$1/\rho_2$	1/1.2
Relative risk aversion	γ_2	1.2
Stock market entry cost	κ_2	15
Type 1 and 2: Preferences		
Discount factor	β_1, β_2	0.91
Preference for housing	ω	0.25
Probability of death	π_t	—
Bequest strength (time periods)	$T_{b,1}, T_{b,2}$	3
Bequest strength (discount factor)	$\bar{\rho}_{b,1}, \bar{\rho}_{b,2}$	0.91
Bequest strength (annuity factor)	$a_{f,1}, a_{f,2}$	0.35

All parameters that govern first or second moments have been annualized. Values in thousands of Swedish kronor (SEK). The standard deviation of the stock market return equals 0.17. The standard deviation of house prices growth equals 0.10. The standard deviation of permanent labor income equals 0.13. The correlation between stock market returns and house price growth equals 0.15. The correlation between stock market returns and idiosyncratic shocks to permanent income equals 0.10. The correlation between house price growth and aggregate permanent income equals 0.63, and the correlation between house price growth and idiosyncratic shocks to permanent income equals 0.14. In addition, the model includes a life-cycle profile for disposable income (g_t). The population weight of type 1 (θ) equals 0.75. See the Online Appendix (Section 2) for details.

5.3 Figure 2: Life-cycle profiles for renters and homeowners

Read.

- Homeowners have higher income than renters at nearly all ages.
- Homeowners' financial wealth exceeds renters' by large margins, especially in middle age.
- Homeowners' stock market participation is high and nearly flat around 80% from age 30 onward.
- Renters' participation remains much lower throughout the life cycle.

Economic message. The gap is a life-cycle pattern, not a one-year accident. Any convincing model needs to match joint dynamics in housing, wealth, and participation.

5.4 Figure 5: Policy rules for type 1 versus type 2

Read.

- Type 2 households consume more and save less at a given level of cash-on-hand.
- The participation threshold is much higher for type 2 than for type 1.
- For type 2 renters, stock market entry becomes worthwhile only near the point where home purchase also becomes attractive.
- Type 2 households therefore accumulate less financial wealth and remain nonparticipants more often.

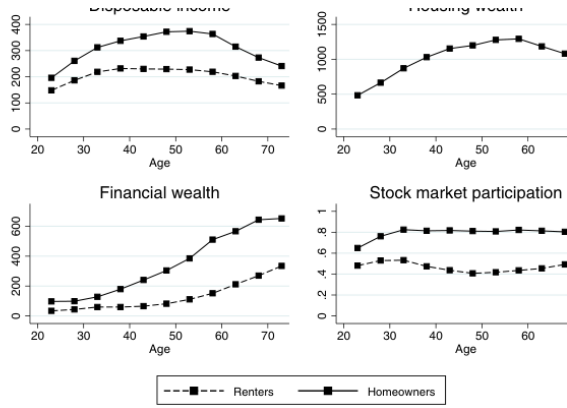


Figure 2
Life-cycle profiles for renters and homeowners
The figure reports the life-cycle profiles for disposable income, housing wealth, financial wealth, and stock participation for renters and homeowners whose household heads have high school education, correspond to the sample of the seventh and eighth columns of Table 2. The dotted lines correspond to renters and lines to homeowners. Statistics are based on five-year age groups starting with group 21–25 and end group 71–75. All statistics are based on wave 2000. The asset values are in terms of thousands of Swedish (SEK).

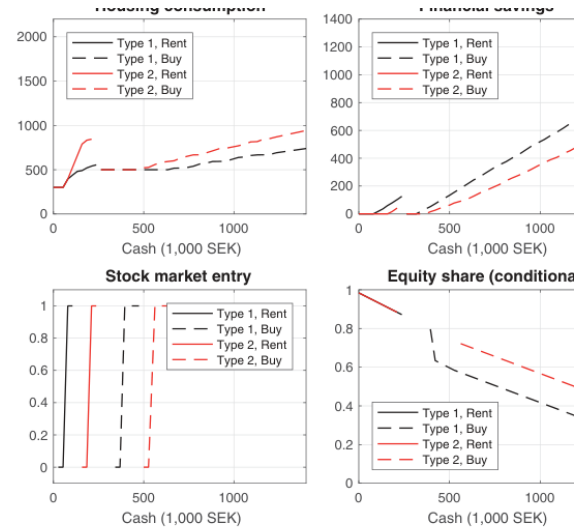


Figure 5
Optimal choice for type 1 and 2
The figure displays policy rules for both types of households. Type 1 has preferences equal to $\beta_1=0$, $\alpha_1=3$ and participation cost (c_{st}) of SEK 5,000. Type 2 has CRRA preferences ($\alpha_2=\alpha_1=3$), $\beta_2=$

Economic message. This figure is the paper’s main mechanism in one picture: heterogeneity in savings motives creates joint sorting into renting and nonparticipation versus owning and participation.

5.5 Table 4: DID estimates in the data

Read.

- For young stable households, a home purchase raises participation by roughly 5 percentage points.
- For young households who both purchase and marry, the participation increase is even larger, around 7 percentage points.
- Stable home sales reduce participation by a few percentage points; sales combined with becoming single reduce participation much more.
- Unconditional equity-share responses are smaller than participation responses, suggesting little movement in conditional equity shares.

Economic message. The event-study evidence does not support the idea that homeownership crowds out participation on the extensive margin. If anything, the direction goes the other way.

5.6 Figure 7: Model fit for homeownership and conditional wealth profiles

Read.

- The model overpredicts homeownership among the young.
- It undershoots housing wealth for homeowners, partly because too many relatively poor households buy homes in the model.
- It matches disposable income and especially financial wealth conditional on tenure reasonably well.

Table 4

Difference-in-difference estimates on equity exposure (Data)

	(1) Partic.	(2) Equity share	(3) Partic.	(4) Equity share	(5) Partic.	(6) Equity share	(7) Partic.	(8) Equity share
Panel A: Home purchases								
Two years before	-0.009 (0.006)	0.018*** (0.006)	0.015 (0.017)	0.020 (0.018)	-0.006 (0.007)	0.006 (0.007)	0.009 (0.028)	0.004 (0.031)
One year before	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Year of purchase	0.052*** (0.005)	0.006 (0.006)	0.070*** (0.016)	0.016 (0.017)	0.006 (0.006)	0.007 (0.006)	0.098*** (0.031)	0.024 (0.030)
One year after	0.049*** (0.006)	0.001 (0.007)	0.096*** (0.020)	0.033* (0.020)	0.004 (0.007)	0.008 (0.007)	0.058 (0.036)	-0.043 (0.036)
Two years after	0.051*** (0.007)	0.011 (0.007)	0.078*** (0.020)	0.018 (0.021)	0.003 (0.008)	0.004 (0.008)	0.072** (0.035)	-0.034 (0.036)
Three years after	0.063*** (0.007)	0.017** (0.008)	0.085*** (0.021)	0.032 (0.021)	-0.009 (0.009)	-0.001 (0.008)	0.029 (0.036)	-0.064* (0.036)
Age	≤ 40	≤ 40	≤ 40	≤ 40	≥ 41	≥ 41	≥ 41	≥ 41
Household struct.	Stable	Stable	S→M	S→M	Stable	Stable	S→M	S→M
Observations	223,067	223,067	19,565	19,565	476,770	476,770	8,484	8,484
R ²	0.005	0.014	0.062	0.026	0.012	0.027	0.028	0.017
Households	37,202	37,202	3,260	3,260	79,464	79,464	1,414	1,414
Panel B: Home sales								
Two years before	-0.034*** (0.007)	-0.008 (0.007)	-0.069 (0.043)	-0.103** (0.049)	-0.002 (0.005)	-0.009* (0.005)	-0.007 (0.027)	-0.028 (0.028)
One year before	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Year of sale	-0.056*** (0.006)	-0.039*** (0.007)	-0.183*** (0.047)	-0.225*** (0.047)	-0.020*** (0.004)	-0.024*** (0.005)	-0.104*** (0.028)	-0.077*** (0.026)
One year after	-0.045*** (0.007)	-0.039*** (0.008)	-0.196*** (0.052)	-0.177*** (0.050)	-0.024*** (0.005)	-0.010* (0.005)	-0.080*** (0.031)	-0.063** (0.027)
Two years after	-0.041*** (0.008)	-0.032*** (0.008)	-0.200*** (0.051)	-0.168*** (0.053)	-0.025*** (0.006)	-0.022*** (0.006)	-0.107*** (0.032)	-0.074** (0.029)
Three years after	-0.035*** (0.009)	-0.032*** (0.009)	-0.217*** (0.061)	-0.172*** (0.059)	-0.037*** (0.007)	-0.030*** (0.006)	-0.116*** (0.034)	-0.083*** (0.032)
Age	≤ 40	≤ 40	≤ 40	≤ 40	≥ 41	≥ 41	≥ 41	≥ 41
Household struct.	Stable	Stable	M→S	M→S	Stable	Stable	M→S	M→S
Observations	204,673	204,673	4,100	4,100	479,243	479,243	11,844	11,844
R ²	0.004	0.011	0.050	0.053	0.014	0.027	0.052	0.039
Households	34,113	34,113	683	683	79,877	79,877	1,974	1,974

The table presents the effects on liquid financial portfolios around the time of a home purchase (panel A) or home sale (panel B) based on the regression in Equation (19)—that is, coefficients denoted by δ_k . In columns where household structure is indicated S→M, the sample is restricted to households whose heads are single in relative year -1 and married in relative year 2. Similarly, M→S denotes a transition from married to single. All specifications include an intercept, year fixed effects, and household fixed effects. Standard errors are clustered at the household level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

collars that diverge, the estimated magnitudes are much greater (Column 3 and

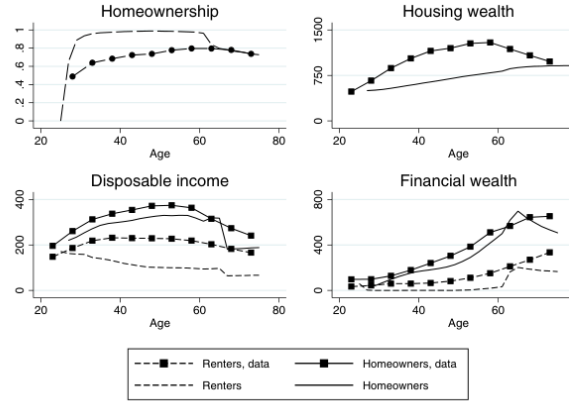


Figure 7

Homeownership and conditional statistics

The top-left panel displays the homeownership rate in the data and the model. The top-right panel displays housing wealth for homeowners. The bottom-left panel displays disposable income for renters and homeowners. The bottom-right panel displays financial wealth for renters and homeowners. Data profiles for housing, disposable income, and financial wealth are reproductions from Figure 2.

Economic message. The model is not perfect on housing margins, but it gets the wealth distribution right well enough to make the participation analysis meaningful.

5.7 Figure 8: Stock market participation in model and data

Read.

- The model matches the overall participation profile well until about age 50, then overshoots late in life.
- For renters versus homeowners, the model produces a large gap in participation during working life.
- Type 1 households alone would not generate the observed gap; type 2 households alone would generate too little participation.
- The combination of the two types reproduces the empirical ranking and a realistic gap size.

Economic message. This is the paper's headline quantitative success: modest heterogeneity plus endogenous tenure choice explains a large part of the homeowner-renter participation gap.

5.8 Table 5: DID estimates in the model

Read.

- Model home purchases strongly raise participation, especially for young buyers.
- Upgrades have little effect on participation but can affect the equity share.

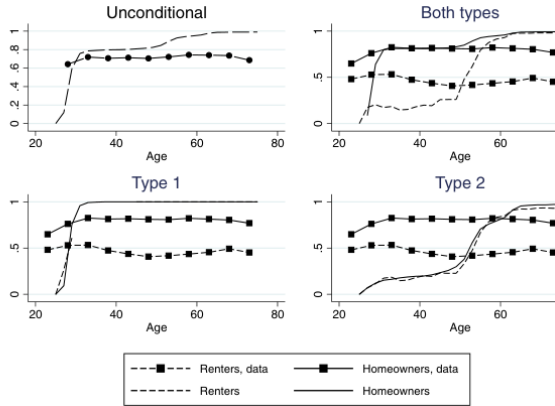


Figure 8
Stock market participation
The top panels are based on types 1 and 2 combined. The top-left panel displays the stock market part rate for renters and homeowners together (i.e., unconditional). The top-right panel displays the stock participation rate for renters and homeowners separately. The bottom panels display the stock market part rates for renters and homeowners separately for type 1 (left) and for type 2 (right). The unconditional data is reproduced from Figure 1. The conditional data profiles are reproduced from Figure 2.

- Home sales and downgrades do not generate positive participation responses.
- The model lines up better with the data on participation than on conditional equity-share movements.

Economic message. The model captures the extensive-margin evidence better than the intensive-margin evidence. That mismatch is informative: it points to a missing force in the model for conditional equity shares.

5.9 Table 6: Other asset classes in the data

	Participation	Equity share (uncond.)	Stocks	Equity funds	Bonds	Money market funds	Bar acct.
Panel A: Home purchases (stable household structure)							
Younger than 40 years	0.054*** (0.005)	—	—	0.046*** (0.006)	0.029*** (0.005)	—	—
Older than 40 years	0.002 (0.006)	—	0.012** (0.005)	0.001 (0.006)	0.009 (0.006)	0.002 (0.006)	—
Panel B: Home purchases (single to married)							
Younger than 40 years	0.083*** (0.017)	−0.020 (0.018)	0.071*** (0.016)	0.079*** (0.019)	0.051*** (0.016)	0.034*** (0.015)	0.08 (0.02)
Older than 40 years	0.071** (0.030)	−0.057* (0.034)	0.112*** (0.033)	0.044 (0.031)	0.046* (0.027)	0.031 (0.028)	0.11 (0.02)
Panel C: Home sales (stable household structure)							
Younger than 40 years	—	−0.037*** (0.006)	—	—	—	—	—
Older than 40 years	−0.027*** (0.005)	—	−0.003 (0.004)	−0.024*** (0.005)	0.007 (0.005)	—	0.01 (0.00)
Panel D: Home sales (married to single)							
Younger than 40 years	−0.201*** (0.045)	—	−0.032 (0.034)	−0.215*** (0.050)	0.041 (0.034)	−0.015 (0.023)	−0.04 (0.02)
Older than 40 years	−0.196*** (0.027)	−0.077*** (0.025)	−0.008 (0.021)	−0.070** (0.028)	−0.018 (0.021)	−0.032 (0.022)	0.00 (0.02)

The table presents the effects of home purchases (panels A and B) and home sales (panels C and D) on major asset classes. Each cell corresponds to a separate regression that is a modified version of Equation (19) where relative year $t=0$ to relative year $t=3$ have been collapsed to a single period. Panels A and C restrict the sample to households whose heads are either single or married both before and after the transaction. Panel B restricts the sample to households whose heads are single before married after the transaction. Panel D restricts the sample to households whose heads are married before but single after the transaction. All regressions include year effects and household fixed effects. Coefficients from regressions that do not fulfill the assumption of parallel trends prior to the transaction (i.e., rejection of $\Delta_{-2}=0$ at $p \leq 0.05$) are omitted and reported as —. Standard errors are clustered at the household level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Read.

- Home purchases are associated with increases in participation across a broader set of risky assets.

	(1) Partic.	(2) Equity share	(3) Partic.	(4) Equity share	(5) Partic.	(6) Equity share
Panel A: Home purchases and upgrades						
Two periods before	−0.141*** (0.049)	−0.000 (0.000)	−0.160*** (0.016)	−0.158*** (0.016)	−0.010 (0.021)	0.010 (0.025)
One period before	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Period of transaction	0.238*** (0.017)	0.001 (0.001)	−0.031*** (0.007)	−0.014* (0.008)	−0.025 (0.020)	0.019 (0.025)
One period after	0.348*** (0.050)	0.364*** (0.015)	−0.019*** (0.007)	0.000 (0.007)	−0.006 (0.020)	0.042* (0.024)
Two periods after	0.380*** (0.050)	0.396*** (0.014)	−0.021*** (0.007)	0.004 (0.008)	−0.010 (0.019)	0.045* (0.023)
Age	25–40	25–40	25–40	25–40	41–75	41–75
Purchases	Yes	Yes	No	No	No	No
Upgrades	No	No	Yes	Yes	Yes	Yes
Observations	3,400	3,400	10,625	10,625	3,615	3,615
R ²	0.427	0.509	0.114	0.109	0.018	0.005
Households	680	680	1,192	1,192	507	507
Panel B: Home sales and downgrades						
Two periods before	−0.021*** (0.006)	0.007 (0.005)	0.049 (0.035)	0.002 (0.030)	0.006 (0.005)	−0.234*** (0.008)
One period before	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)	0.000 (.)
Period of transaction	−0.114 (0.083)	0.010 (0.033)	−0.062* (0.037)	−0.108*** (0.035)	0.009*** (0.003)	−0.221*** (0.006)
One period after	−0.110 (0.083)	0.024 (0.028)	−0.099** (0.038)	−0.161*** (0.039)	0.006* (0.003)	−0.216*** (0.006)
Two periods after	−0.108 (0.083)	0.032 (0.025)	−0.222*** (0.050)	−0.311*** (0.049)	0.000 (0.004)	−0.215*** (0.007)
Age	41–75	41–75	25–40	25–40	41–75	41–75
Sales	Yes	Yes	No	No	No	No
Downgrades	No	No	Yes	Yes	Yes	Yes
Observations	2,695	2,695	815	815	8,860	8,860
R ²	0.038	0.029	0.110	0.195	0.011	0.380
Households	539	539	82	82	1,309	1,309

The table presents the effects of home purchases and home upgrades (panel A) and home sales and home downgrades (panel B) in the model, based on the regression specified in Equation (19). All specifications include household fixed effects and an intercept which is not reported. Standard errors are clustered at the household level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

- Home sales are associated with reductions in risky-asset exposure, especially when combined with divorce.
- The broad-asset evidence reinforces the main result from stock-market participation alone.

Economic message. The non-crowding-out message is not confined to one narrow participation definition. It generalizes to a broader risky-asset portfolio.

6 Short summary

- The paper studies why homeowners participate in the stock market much more than renters.
- A pure crowding-out story predicts the wrong cross-sectional ranking: otherwise identical renters should participate more than otherwise identical homeowners.
- The paper resolves this by combining endogenous housing tenure with preference heterogeneity.
- High-savings households accumulate more financial wealth, become homeowners, and pay the stock-market entry cost.
- Low-savings households remain poorer, rent more often, and often stay out of the stock market.
- In the data, home purchases do not reduce participation, and home sales do not raise it.
- In the model, the participation gap between homeowners and renters is quantitatively close to the data, especially during working life.
- The model performs best on the extensive margin and less well on conditional equity shares.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the large gap in stock market participation between renters and homeowners need not reflect a direct positive causal effect of homeownership on participation. Instead, the gap can arise because households differ in their savings motives, those motives shape wealth accumulation, and wealth accumulation shapes both tenure choice and stock market entry.

7.2 Economic intuition

Housing is a competing use of wealth. For a given household, owning a home can indeed crowd out stock market participation. But equilibrium cross sections are shaped by sorting. Households who save a lot end up both owning homes and participating in the stock market. Households who save little tend to remain renters and nonparticipants. That sorting force can dominate the within-household crowding-out force.

7.3 Takeaway

The paper makes a precise and important point: one should not interpret the homeowner-renter participation gap as evidence that housing mechanically crowds out equity investment. A substantial part of the gap reflects heterogeneity in household types and the joint determination of housing and portfolio choice over the life cycle.